

Create a linkedlist with the node having data part & pointer part where data is a character and pointer points to its next node. Accept a string where the character part of each string is stored as the data part in the node now find out if the linked is palindrome or not.

```
#include <stdio.h>
```

```
struct node
```

```
{
```

```
    char data;
```

```
    struct node next;
```

```
};
```

```
struct node *head
```

```
void struct node* createNode (char ch)
```

```
{
```

```
    struct node * newNode = (struct node*) malloc  
        (sizeof (struct node));
```

```
    newNode -> data = ch;
```

```
    newNode -> next = NULL;
```

```
    return newNode;
```

```
}
```

```
int isPalindrome (struct node *head, int length)
```

```
{
```

```
    char str[100];
```

```
    int i=0;
```

```
    while (head != NULL)
```

```
    {
```

```
        str[i++] = head -> data;
```

```
        head = head -> next;
```

```
    }
```

```
    for (i=0; i < length/2; i++)
```

```
    { if (str[i] != str[length-i-1])
```

```
        { return 0;
```

```
    }
```

```

    return 1;
}

int main()
{
    struct node * head = NULL, * temp = NULL, * newnode;
    char str[100];
    int i, length;
    printf("Enter a string:");
    scanf("%s", &str);
    length = strlen(str);
    for(i=0; i<length; i++)
    {
        newNode = createNode(str[i]);
        if(head == NULL)
        {
            head = newNode;
            temp = newNode;
        }
        else
        {
            temp->next = newNode;
            temp = newNode;
        }
    }

    printf("Linked list:");
    temp = head;
    while(temp != NULL)
    {
        printf("%c->", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
    if(isPalindrome(head, length))
        printf("The linked list is a palindrome.\n");
    else
        printf("The linked list is not a palindrome.\n");
    return 0;
}

```